# ReplicationPad3d#

https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad3d.html

**Description:**

Pads the input tensor using replication of the input boundary. For N-dimensional padding, use torch.nn.functional.pad(). padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom, padding_front\text{padding_front}padding_front, padding_back\text{padding_back}padding_back) Note that the output dimensions must remain positive. Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win). Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

## Function Signature

```
class torch.nn.ReplicationPad3d(padding)[source]#
Parameters
Shape:
```

## Code Examples

```
>>> m = nn.ReplicationPad3d(3)
>>> input = torch.randn(16, 3, 8, 320, 480)
>>> output = m(input)
>>> # using different paddings for different sides
>>> m = nn.ReplicationPad3d((3, 3, 6, 6, 1, 1))
>>> output = m(input)
```

## Detailed Documentation

### Documentation

Pads the input tensor using replication of the input boundary.

For N-dimensional padding, use torch.nn.functional.pad().

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (padding_left\text{padding_left}padding_left, padding_right\text{padding_right}padding_right, padding_top\text{padding_top}padding_top, padding_bottom\text{padding_bottom}padding_bottom, padding_front\text{padding_front}padding_front, padding_back\text{padding_back}padding_back) Note that the output dimensions must remain positive.

Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where

Dout=Din+padding_front+padding_backD_{out} = D_{in} + \text{padding_front} + \text{padding_back}Dout=Din+padding_front+padding_back

Hout=Hin+padding_top+padding_bottomH_{out} = H_{in} + \text{padding_top} + \text{padding_bottom}Hout=Hin+padding_top+padding_bottom

Wout=Win+padding_left+padding_rightW_{out} = W_{in} + \text{padding_left} + \text{padding_right}Wout=Win+padding_left+padding_right